

AIT 511 Assignment Polynomial Regression

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Code Repository: github.com/RohanCoderiiitb/AIT-511-Machine-Learning-Assignment-1

1. Overall Approach

The assignment asks for polynomial-regression models for two personalized continuous-response problems, with test performance evaluated by MSE and R^2 . The important modeling issue is not simply to obtain a low training error: polynomial expansion rapidly enlarges the hypothesis space, so a degree that fits the training data extremely well can generalize poorly. The notebooks therefore treat polynomial degree and regularization as model-selection variables and use the training set only for selection, leaving the test set untouched until inference.

The implemented pipeline is: **(1) data audit** → **(2) relationship and distribution analysis** → **(3) total-degree polynomial expansion** → **(4) fold-wise scaling** → **(5) ridge/Lasso/target-scale investigation where relevant** → **(6) joint degree-regularization cross-validation** → **(7) refit on all training data** → **(8) prediction-file validation**. Each stage answers a different question: whether the data are usable, what structure the model must represent, how large the hypothesis class should be, how to stabilize it, and whether the selected model remains predictive on unseen training folds.

Polynomial representation

For p input variables and total degree d , the notebook constructs every monomial whose total exponent sum is at most d , excluding the constant term. The number of columns is therefore

$$P(d, p) = \binom{d+p}{p} - 1.$$

This is important because the expansion contains both nonlinear marginal effects ($x_i^2, x_i^3, \dots$) and interactions ($x_i x_j, x_i^2 x_j, \dots$). It does *not* assume that the effect of one variable is independent of the others. Problem 1 has 461 terms at degree 5 and 8007 at degree 10; Problem 2 has 285 terms at degree 10 and 1770 at degree 20. The increasing dimension explains why degree selection and regularization must be treated together.

Scaling and regularization

The expanded columns can have very different numerical scales, especially at higher degree. The notebooks standardize the polynomial columns before fitting. Crucially, in every CV fold the scaler is fitted only on the fold's training portion and then applied to its validation portion; this prevents information from the validation fold from entering the preprocessing step. The fitted model uses ridge regression,

$$\min_{\mathbf{w}, b} \frac{1}{n} \|\mathbf{y} - X\mathbf{w} - b\mathbf{1}\|_2^2 + \lambda \|\mathbf{w}\|_2^2,$$

with the intercept unpenalized. In centered coordinates,

$$\mathbf{w} = \left(\frac{X_c^T X_c}{n} + \lambda I \right)^{-1} \frac{X_c^T y_c}{n}, \quad b = \bar{y} - \bar{\mathbf{x}}^T \mathbf{w}.$$

Ridge is especially appropriate here because polynomial expansion creates correlated basis columns and potentially ill-conditioned directions. The penalty shrinks unstable coefficients while retaining the interaction structure needed to model nonlinear response surfaces.

Model selection and validation logic

For each candidate degree, the notebooks search a logarithmic grid of λ values and compute 5-fold validation MSE. The final pair (d, λ) is the global minimum over the two-dimensional CV surface, rather than a degree selected from training error. This matters because increasing degree almost always decreases training error; only validation error can reveal when the added flexibility has stopped generalizing. R^2 is reported as a secondary measure of explained variance, while MSE is the primary selection criterion because it is directly aligned with the assignment's evaluation metric.

Feature correlation is deliberately not used as a feature-selection rule. Pairwise correlation measures only linear association. A variable with weak marginal correlation can still contribute strongly through x_i^2 or an interaction, while a strongly correlated variable need not improve the multivariate polynomial model. The notebooks instead use correlation matrices, feature-target plots, train/test moments, and boundary statistics as diagnostics, and let held-out prediction error decide whether the corresponding polynomial flexibility is useful.

2. Problem 1: Power Plant Steam Turbine Optimization

2.1 Data Inspection

The first problem models the Net Power Score of a multi-stage steam turbine from six operational parameters. There are 1000 training rows and 1000 test rows. All seven training columns are numeric and complete; there are no missing values and no duplicate training rows. The six inputs are bounded in $[-1, 1]$ and show substantial mass exactly at the boundaries. In the training set, approximately 29–33% of each feature is at either -1 or $+1$; in the test set this rises to approximately 47–52%. The boundary values were retained because they are part of the supplied input distribution rather than missing-value artifacts.

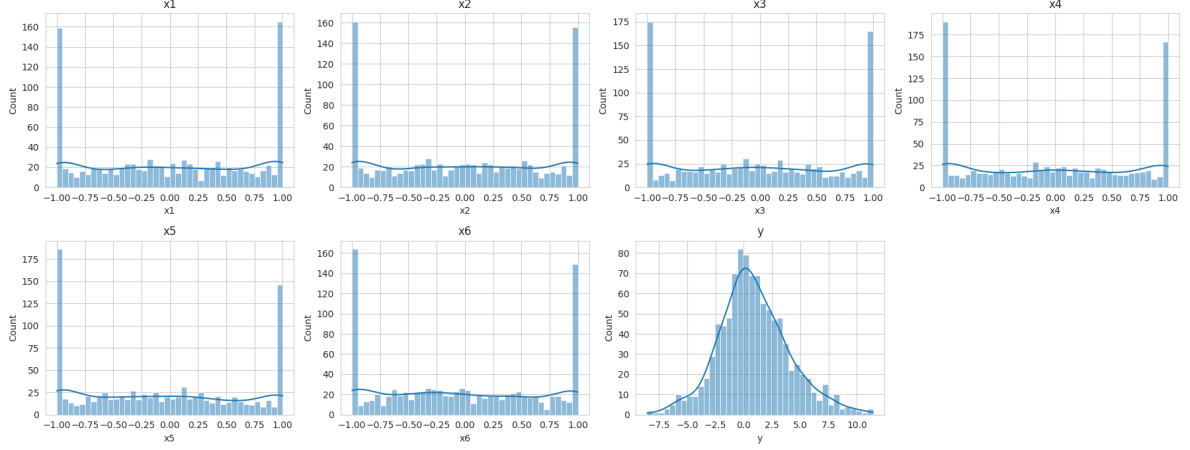


Figure 1: Problem 1 training distributions of the six operational variables and Net Power Score.

The target has mean 0.889, standard deviation 3.120, and range $[-8.544, 11.422]$. Twenty-five observations lie outside the IQR-based target fences. They were not removed: the objective is to learn the supplied response surface, and treating statistically unusual responses as errors without evidence would change the learning problem.

2.2 Feature and Target Analysis

The feature–target plots are individually noisy: there is no single predictor whose raw scatterplot explains the Net Power Score well. The training correlation matrix makes this quantitative: the correlations of y with $x_1, x_2, x_3, x_4, x_5, x_6$ are approximately 0.03, 0.03, 0.04, 0.01, 0.38, -0.21 , respectively. Thus x_5 has the strongest marginal linear association, but even it explains only a limited part of the response variation by itself. This is important because a weak marginal correlation does not imply that a variable is irrelevant to a multivariate nonlinear model. Effects can appear through squared terms, sign-dependent curvature, or interactions such as $x_i x_j$. The appropriate response is therefore not to discard the weakly correlated variables, but to give the polynomial model access to joint terms and let cross-validation determine whether the added capacity improves prediction.

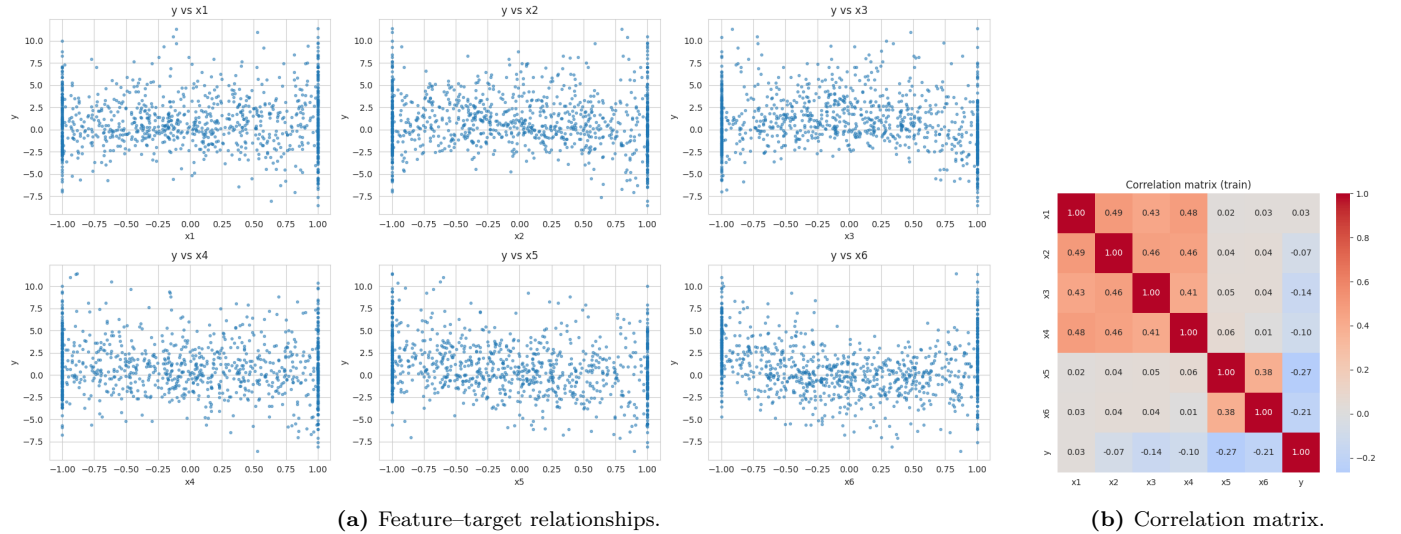


Figure 2: Problem 1 relationship diagnostics.

The predictor correlation matrix is used to check linear redundancy and to assess whether the dependence structure changes between train and test. The maximum absolute difference between corresponding entries of the train and test feature-correlation matrices is 0.093. This is a modest shift relative to a complete change of dependence structure, so

there is no strong evidence of catastrophic pairwise covariate shift. However, it is not appropriate to conclude from this number alone that the model will generalize: polynomial terms can create dependencies that are invisible in the raw correlation matrix.

The more consequential distributional difference is the boundary concentration. The notebook explicitly stratifies out-of-fold error by the number of clipped/boundary-valued features. Training rows with 4 or more clipped features constitute only 9.1% of training data but 32.2% of test data; their OOF MSE is 0.944, compared with 0.386 for rows with no clipped features. Reweighting the OOF errors to the test-set mixture increases the estimated MSE from 0.5174 to 0.6666. This is an important robustness finding: the overall CV score is strong, but the test distribution places substantially more mass in a region where validation error is higher. The final model is therefore selected by ordinary CV, while this stratified analysis provides a realistic caution about expected test difficulty.

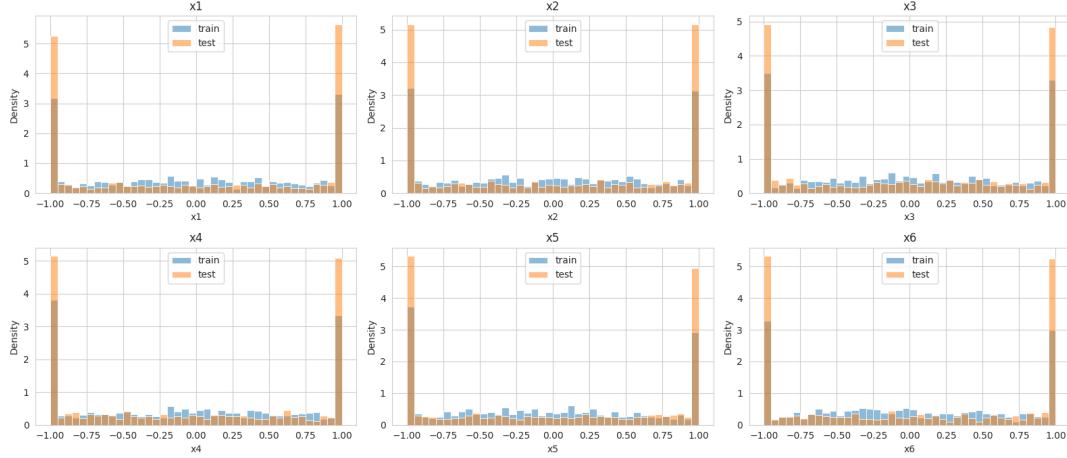


Figure 3: Train/test feature distributions. The wider test spread and stronger boundary mass motivate the explicit distribution-shift check.

2.3 Polynomial Model Selection

The search covers degrees 1–10 and 25 ridge values from 10^{-4} to 10^2 . For each degree, the best λ is found by 5-fold CV, after which the best degree– λ combination is selected globally. The progression of the best CV MSE by degree is informative: degree 1 gives 8.6479, degree 2 gives 2.6781, degree 3 gives 0.8913, degree 4 gives 0.7016, and degree 5 reaches 0.5174. After that, the error increases to 0.6021, 0.6319, 0.7546, 0.8839 and 1.0017 for degrees 6–10. Thus, the additional nonlinear and interaction terms up to degree 5 capture reproducible structure; beyond degree 5, the rapidly growing basis introduces more variance than useful signal.

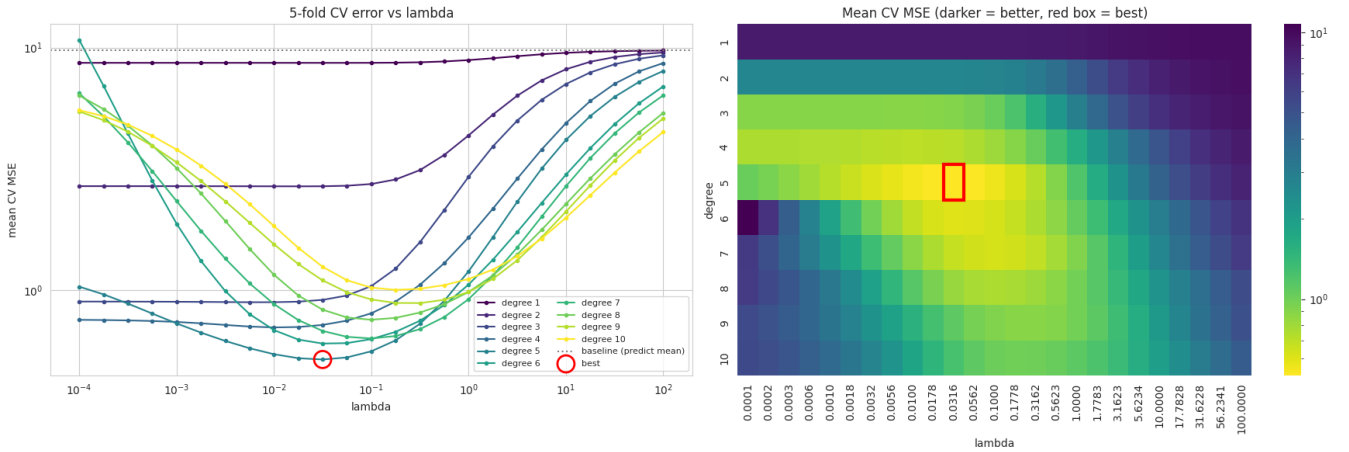


Figure 4: Problem 1 5-fold CV error across ridge strength and polynomial degree. The red marker identifies the global validation optimum.

2.4 Final Model and Validation

Chosen model: all 6 features | degree 5 | $\lambda = 0.03162$ | 461 terms
CV MSE: 0.5174 ± 0.0959 | training MSE 0.2019 | training $R^2 = 0.9792$

The selected model is refit on all 1000 training observations. Its 5-fold out-of-fold MSE is again 0.5174, confirming that the reported validation score corresponds to genuinely held-out predictions. An independent 800/200 hold-out

check gives MSE 0.5072 and $R^2 = 0.9501$. The notebook also verifies that the final closed-form ridge solution and the gradient-descent implementation produce test predictions differing by only 5.1×10^{-13} . The final CSV has 1000 rows, no NaNs, and exactly preserves the supplied test inputs.

3. Problem 2: Subterranean Thermal Reservoir Mapping

3.1 Data Inspection

The second problem predicts a Thermal Anomaly Score from three spatial coordinates: east–west offset, north–south offset, and vertical depth offset. There are 1000 training and 1000 test rows, with no missing values and no duplicate training rows. Boundary values are again common: roughly 10.7–13.1% of training observations lie at each individual lower/upper boundary depending on the feature, with similar proportions in the test set. Unlike Problem 1, the aggregate train/test boundary mixture is very close: 41.3% of training rows have no clipped features versus 44.4% in test, and only about 1% of training rows and 1.2% of test rows have all three features clipped.

There are six repeated spatial coordinates in training. Their pooled within-coordinate target variance is only 0.195, compared with an overall target variance of 40.889. This is consistent with a largely deterministic spatial response and does not suggest that repeated coordinates introduce substantial random measurement noise. The target has mean 2.319, standard deviation 6.394, and range $[-29.328, 38.966]$; 83 points lie outside the IQR fences. These observations are retained because no evidence in the assignment establishes them as erroneous.

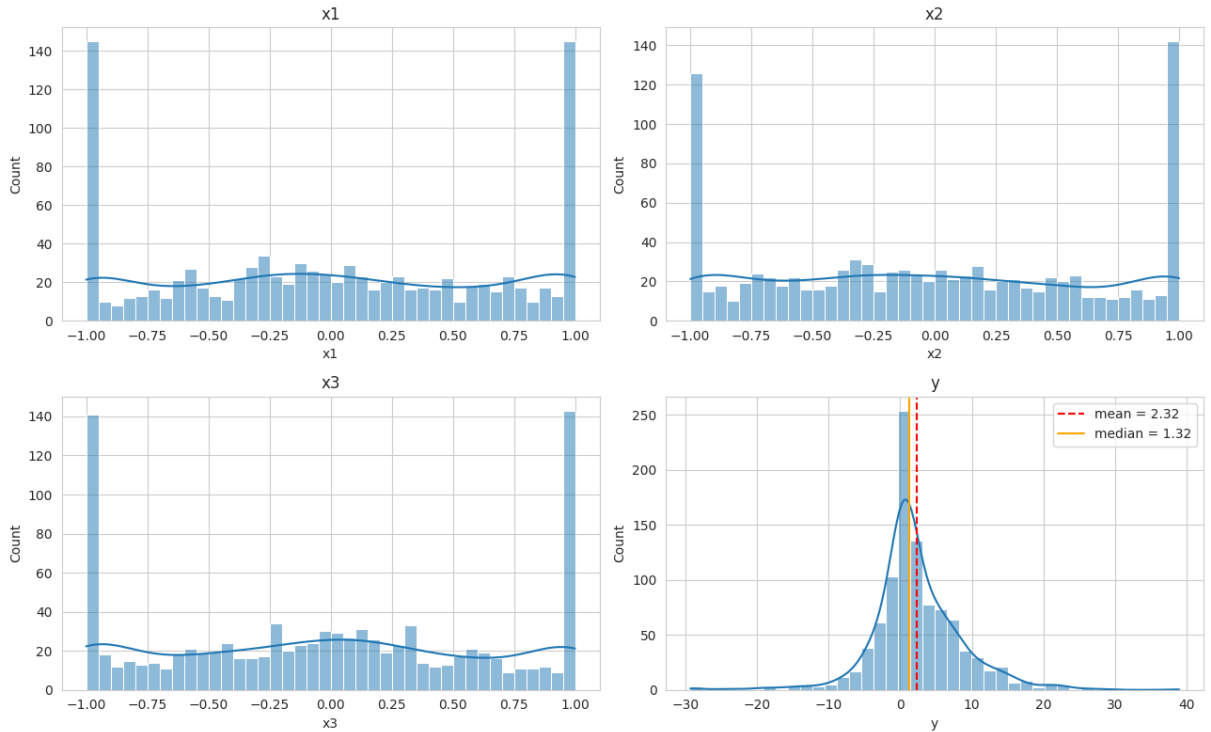


Figure 5: Problem 2 training distributions of the three spatial variables and the Thermal Anomaly Score. The output distribution is positively skewed with substantial tails, while the input variables exhibit pronounced boundary mass.

3.2 Feature and Target Analysis

The feature–target plots show a response that is concentrated near the center of the target range but develops strong tails and boundary-dependent behavior, particularly toward the edges of the coordinate domain. A linear model would constrain the response to a plane in (x_1, x_2, x_3) and therefore could not represent such curvature or coordinate interactions. The polynomial basis is designed to capture exactly this structure: it can combine marginal powers with terms such as x_1x_2 , $x_2^2x_3$, and higher-order interactions. Because the three raw coordinates have similar scale and very stable train/test dependence, retaining all three and increasing the polynomial degree is a more defensible modeling strategy than selecting variables from pairwise correlation alone.

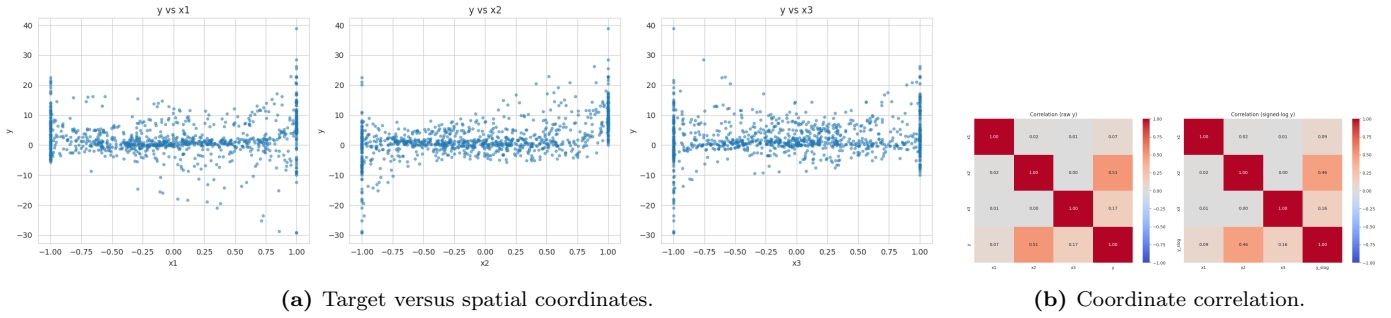


Figure 6: Problem 2 feature-relation diagnostics.

The maximum absolute difference between train and test feature-correlation matrices is only 0.017, substantially smaller than the corresponding Problem 1 value of 0.093. Thus the three-coordinate dependence structure is highly stable between splits. This supports using the training CV results as a useful estimate of test behavior, although it still cannot replace actual hidden test evaluation.

The correlation matrix should not be interpreted as evidence that only highly correlated variables matter. The polynomial model is specifically intended to learn nonlinear and interaction effects, and raw pairwise correlation is blind to many such relationships. Consequently, all three coordinates are retained and the degree is determined from held-out MSE.

3.3 Polynomial Model Selection

Problem 2 additionally investigates whether transforming the target improves numerical behavior or prediction on the original MSE scale. The signed-log transform

$$y_{\log} = \text{sign}(y) \log(1 + |y|)$$

is exactly invertible for the supplied targets. The notebook compares Ridge, Lasso and unregularized least squares over polynomial degrees. On the original target scale, the best signed-log model has CV MSE 2.7640, whereas the best raw-target model achieves 0.2476. Therefore the transform is rejected for the final model under the assignment’s metric. This is an important methodological decision: a transformation is not adopted merely because it compresses the target; it must improve prediction after transforming the model output back to the metric’s original scale.

The comparison also demonstrates the need for regularization. Raw-target least squares has CV MSE 0.446 at degree 10 but explodes to 2.88×10^{13} at degree 15; the signed-log sweep shows the same high-degree instability. Ridge and Lasso suppress these unstable coefficient directions, with Ridge giving the lowest CV MSE at degree 10.

The final degree search covers 1–20 and ridge penalties from 10^{-6} to 10^2 . The best validation MSE decreases from 29.179 at degree 1 to 3.767 at degree 4, 1.463 at degree 5, 0.578 at degree 6, 0.340 at degree 7, 0.262 at degree 8, 0.260 at degree 9, and reaches 0.2476 at degree 10. It then rises gradually through degree 20. This pattern is strong evidence that degree 10 captures the required spatial complexity while degrees above 10 mainly add variance.

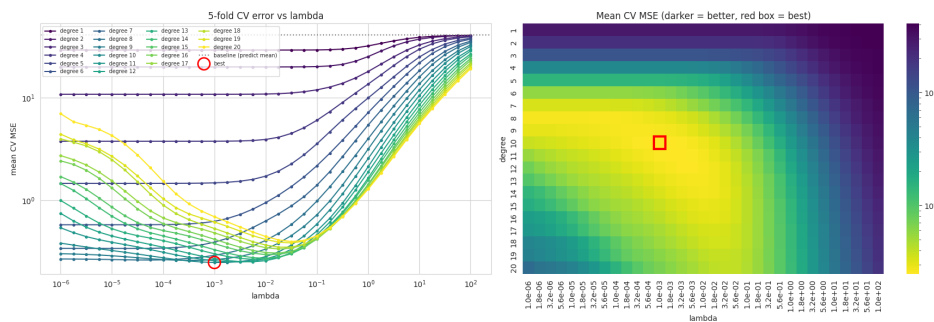


Figure 7: Problem 2 5-fold CV surface over degree and ridge strength.

3.4 Final Model and Validation

Chosen model: all 3 features | degree 10 | $\lambda = 10^{-3}$ | 285 terms
CV MSE: 0.2476 ± 0.0057 | training MSE 0.1683 | training $R^2 = 0.9959$

The five validation folds give MSEs 0.2419, 0.2427, 0.2555, 0.2510 and 0.2469, indicating low fold-to-fold variation. Mean fold-wise R^2 is 0.9937 and pooled out-of-fold R^2 is 0.9939. The small training/OOF gap (0.1683 versus 0.2476) is consistent with a high-capacity model that is nevertheless regularized enough to generalize. The final prediction CSV contains 1000 rows, zero NaNs, and the original test coordinates are preserved exactly.

4. Conclusion

The final models are not selected by visual fit, training error, or raw feature correlation. The notebooks establish a complete validation argument: the supplied data are audited; train/test distributions and dependence structures are compared; feature–target plots establish the need for nonlinear terms; polynomial expansion provides powers and interactions; fold-wise standardization prevents preprocessing leakage; ridge/Lasso/target-scale alternatives are investigated where relevant; and the degree and regularization strength are selected from 5-fold validation MSE. The final models are then refit on all training data and their prediction files are structurally checked.

The two problems require different effective complexity. For **Power Plant Steam Turbine Optimization**, degree 5 is optimal under CV, but the test distribution contains substantially more boundary-heavy rows, so the notebook’s stratified OOF analysis identifies a realistic increase in expected error under the test mixture. For **Subterranean Thermal Reservoir Mapping**, the train/test dependence structure is much more stable, the raw target clearly outperforms the signed-log alternative, and degree 10 provides the best validation performance with a very small fold-to-fold spread. These conclusions are directly supported by the executed notebooks and the complete training/inference code is provided in the GitHub repository above.